

Change of basis equations

N.J. Theron
March 2026

1 Points or vectors (1st order tensors)

Consider two frames A and B , as shown in MLS fig. 2.1 (p. 23). Frame B is obtained through a rotation defined by the rotation matrix R_{ab} from frame A , exactly as in MLS section 2. R_{ab} can also be used to describe the rotation of a vector $\vec{v}$ to form a new vector $\vec{w}$, where the components of these vectors in frame A can be expressed as column vectors v and w , respectively. It can be shown that $w = R_{ab}v$. In the same way, R_{ab} can also be used to describe the rotation of a point with coordinates in frame A expressed as the column vector h_a to form a new point with coordinates in frame A expressed as the column vector q_a , where it can be shown that

$$q_a = R_{ab}h_a . \quad (1)$$

Note that h and q are two **different** points.

Let the coordinates of the new point q expressed in frame B be written in the column vector q_b . Since the new point q is obtained by exactly the same rotation that produced frame B , it is however true that the coordinates of the new point q in frame B is the same as those of the original point h in frame A , that is h_a . Therefore, if we wish to calculate the coordinates q_b of the new point q in frame B , this can be achieved by using the inverse of eq. (1):

$$q_b = h_a = R_{ab}^T q_a , \quad (2)$$

The inverse of eq. (2) is

$$q_a = R_{ab}q_b , \quad (3)$$

which is exactly the same and has exactly the same meaning, as the transformation equation (unnumbered) given by MLS in the upper part of p. 25.

Equations (2) and (3) are called the change of orthonormal basis equations for 1st order tensors $\in \mathbb{R}^3$, as it describes the how the coordinates of a single point (that is, **the same point**) or a single vector (that is, **the same vector**) is expressed in two different orthonormal bases or frames.

2 2nd order tensors

Consider two vectors $\vec{v}$ and $\vec{w}$ expressed in component form in frame B (or basis B) as

$$v_b = \begin{bmatrix} v_{b1} \\ v_{b2} \\ v_{b3} \end{bmatrix} \quad w_b = \begin{bmatrix} w_{b1} \\ w_{b2} \\ w_{b3} \end{bmatrix} , \quad (4)$$

where the subscripts (1, 2, 3) are used to indicate the (x, y, z) axes in all frames.

The outer product H_b of the two 1st order tensors w_b and v_b is obtained when the vector w_b is post-multiplied by the transposed of the vector v_b . This outer product is a second order tensor expressed in components in frame B :

$$H_b = w_b v_b^T = \begin{bmatrix} w_{b1} \\ w_{b2} \\ w_{b3} \end{bmatrix} \begin{bmatrix} v_{b1} & v_{b2} & v_{b3} \end{bmatrix} = \begin{bmatrix} w_{b1}v_{b1} & w_{b1}v_{b2} & w_{b1}v_{b3} \\ w_{b2}v_{b1} & w_{b2}v_{b2} & w_{b2}v_{b3} \\ w_{b3}v_{b1} & w_{b3}v_{b2} & w_{b3}v_{b3} \end{bmatrix} . \quad (5)$$

If frame B is obtained by a rotation R_{ab} from frame A , the components of vectors $\vec{v}$ and $\vec{w}$ in frame A can be determined by using the change of basis equation (3) above:

$$v_a = R_{ab}v_b \quad w_a = R_{ab}w_b \quad (6)$$

and hence eq. (5) can be written in terms of v_a and w_a , by using the inverses of eq. (6):

$$H_b = R_{ab}^T w_a v_a^T R_{ab} .$$

Since $H_b = w_b v_b^T$ is the second order tensor expressed in frame B , $w_a v_a^T$ must be the same second order tensor expressed in frame A , $H_a = w_a v_a^T$. The above equation can therefore be written as

$$w_a v_a^T = H_a = R_{ab} H_b R_{ab}^T . \quad (7)$$

From this follows

$$H_b = R_{ab}^T H_a R_{ab} . \quad (8)$$

Eq.s (7) and (8) are the change of basis equations for second order tensors.